Class-Book Misses: Redo Set

Cici Cisneros · SAT Math · 11 questions · November 11, 2026

LINES, SLOPES, FUNCTIONS, SYSTEMS

Cici: these are the eleven questions from your SAT class book that got away. Redo each one here and show every line, signs included. Before you move on, check your answer a second way: plug it back in, graph it in Desmos, or take a different algebra path. The numbers match your Revision Bank on the hub, so you can enter each answer there too.

#19 The line $ax + by = 18$ passes through the points $(2, 3)$ and $(6, 1)$. What is the value of $\frac{a}{b}$?

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{4}$

SHOW EVERY LINE

☐ Checked a second way

#20 When the equation $px - 3y = 12$ is rewritten in the form $y = mx + b$, the slope is -2. What is the value of p?

(A) -6 (B) -2 (C) 2 (D) 6

SHOW EVERY LINE

☐ Checked a second way

#21 The line $ax + by = 18$ is perpendicular to the line $5x - 3y = 12$. What is the value of $\frac{a}{b}$?

(A) $-\frac{3}{5}$ (B) $-\frac{5}{3}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$

SHOW EVERY LINE

☐ Checked a second way

#22 Two lines are given by $y = (2k - 1)x + 5$ and $4x + 2y = 7$. For what value of k are the lines perpendicular?

(A) $-\frac{1}{4}$ (B) 0 (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

SHOW EVERY LINE

☐ Checked a second way

#23 The slope of a line is 3 times its y-intercept. If the line passes through $(2, 20)$, what is the y-intercept?

(A) $\frac{20}{7}$ (B) 4 (C) $\frac{30}{7}$ (D) 5

SHOW EVERY LINE

☐ Checked a second way

#24 The equation of a line is written in slope-intercept form $y = mx + b$. If increasing x by 4 increases y by 10, and the line passes through $(2, 7)$, what is the value of b?

(A) -1 (B) 0 (C) 2 (D) 5

SHOW EVERY LINE

☐ Checked a second way

#25 A function f is linear and satisfies $f(x + 3) - f(x - 1) = 20$ for all real x. If $f(2) = 7$, what is $f(6)$?

(A) 17 (B) 21 (C) 27 (D) 31

SHOW EVERY LINE

☐ Checked a second way

#26 An object is dropped from a cliff. The distance, in meters, that it has fallen after t seconds is modeled by the function $d(t) = 4.9t^2$. What is the average speed (average rate of change) over the interval from $t = 1$ to $t = 3$?

(A) 4.9 meters per second
(B) 9.8 meters per second
(C) 19.6 meters per second
(D) 39.2 meters per second

SHOW EVERY LINE

☐ Checked a second way

#27 The average rate of change of a function $f(x) = x^2 + kx$ from $x = 3$ to $x = 7$ is 20. What is the value of k?

Ⓐ 4 Ⓑ 6 Ⓒ 8 Ⓓ 10

SHOW EVERY LINE

☐ Checked a second way

#28 Consider the system of equations:

$$4x + 2y = 18$$
$$8x + 4y = 36$$

For each real number r, which of the following points lies on the graph of both equations in the xy-plane?

Ⓐ $(r,\ 2r - 9)$ Ⓑ $(r,\ 9 - 2r)$ Ⓒ $(r,\ 18 - 2r)$ Ⓓ $(9 - 2r,\ r)$

SHOW EVERY LINE

☐ Checked a second way

#29 Tickets for a movie cost \$8.00 for adults and \$5.00 for children. A total of 200 tickets were sold, and the total revenue was \$1,360. How many adult tickets were sold?

Answer: _____

SHOW EVERY LINE

☐ Checked a second way